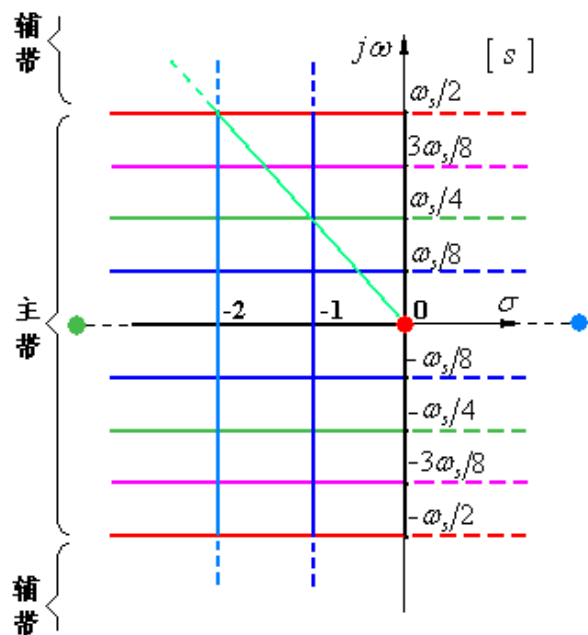
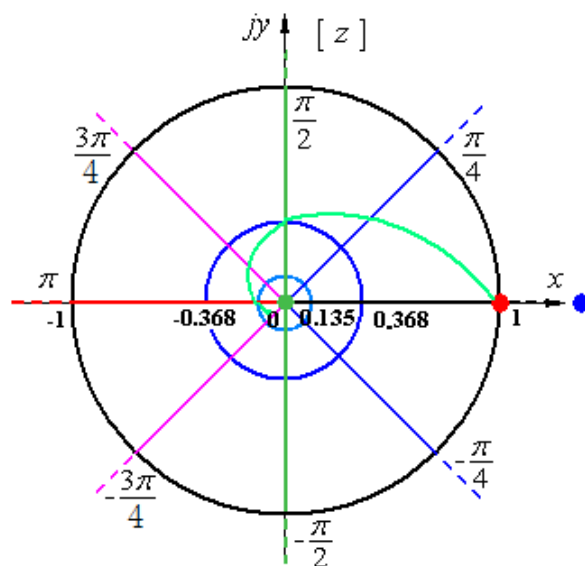




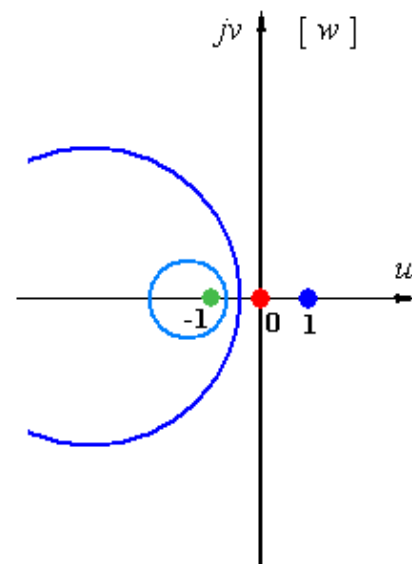
$$s = \sigma + j\omega$$

$$\begin{cases} z = e^{Ts} = e^{(\sigma + j\omega)T} \\ = e^{\sigma T} \cdot e^{j\omega T} = |z| \angle z \end{cases}$$



$$z = x + jy$$

$$\begin{cases} |z| = e^{\sigma T} \\ \angle z = \omega T \end{cases}$$



$$w = u + jv$$

$$\begin{cases} z = \frac{1+w}{1-w} \\ w = \frac{z-1}{z+1} \end{cases}$$

$$\left\{ \begin{array}{l} \sigma = \begin{cases} -\infty \\ -1 \\ 0 \\ 1 \\ \infty \end{cases}, \omega \text{ 任意} \quad Ts = \begin{cases} -\infty T + j\omega T \\ -T + j\omega T \\ 0T + j\omega T \\ T + j\omega T \\ \infty T + j\omega T \end{cases} \quad (\text{垂直线}) \\ \\ \sigma \text{ 任意}, \omega = \begin{cases} 0 \\ \omega_s/8 \\ \omega_s/4 \\ 3\omega_s/8 \\ \omega_s/2 \end{cases} \quad Ts = \begin{cases} \sigma T + j0T \\ \sigma T + j\omega_s T/8 \\ \sigma T + j\omega_s T/4 \\ \sigma T + j3\omega_s T/8 \\ \sigma T + j\omega_s T/2 \end{cases} \quad (\text{水平线}) \end{array} \right.$$

$$|z| = \begin{cases} e^{-\infty T} = 0 \\ e^{-T} \xrightarrow{T=1} 0.368 \\ e^{0T} = 1 \\ e^T \xrightarrow{T=1} 2.718 \\ e^{\infty T} = \infty \end{cases} \quad (\text{同心圆})$$

$$\angle z = \omega T = \omega \frac{2\pi}{\omega_s} = \begin{cases} 0 \\ \pi/4 \\ \pi/2 \\ 3\pi/4 \\ \pi \end{cases} \quad (\text{射线})$$

§ 6.5 离散系统的稳定性与稳态误差

§ 6.5.1 $s \rightarrow z$ 映射

§ 6.5.2 离散系统稳定的充要条件:

$\Phi(z)$ 的全部极点均位于 z 平面的单位圆内

证明:

$$\Phi(z) = \frac{M(z)}{D(z)} = \frac{\prod_{i=1}^m (z - \alpha_i)}{\prod_{j=1}^n (z - \beta_j)} = \sum_{j=1}^n \frac{C_j z}{z - \beta_j} = K(z)$$

$$c(k) = \sum_{j=1}^n C_j \beta_j^k \xrightarrow{k \rightarrow \infty} 0 \Rightarrow |\beta_j| < 1$$

$$c^*(t) = \sum_{k=0}^{\infty} \left(\sum_{j=1}^n C_j \beta_j^k \right) \cdot \delta(t - kT)$$

§ 6.5.3 离散系统的稳定判据

1. w 变换及 w 域的劳斯稳定判据

(1) w 变换:

$$\begin{cases} z = \frac{w+1}{w-1} \\ w = \frac{z+1}{z-1} \end{cases} \quad \begin{cases} z = \frac{1+w}{1-w} \\ w = \frac{z-1}{z+1} \end{cases} \quad \begin{cases} z = \frac{1+\frac{T}{2}w}{1-\frac{T}{2}w} \\ w = \frac{2}{T} \cdot \frac{z-1}{z+1} \end{cases}$$

设 $z = x + jy$, $w = u + jv$

$$w = \frac{z+1}{z-1} = \frac{x+1+jy}{x-1+jy} = \frac{(x+1+jy)(x-1-jy)}{(x-1)^2 + y^2}$$

则

$$= \frac{x^2 - 1 + y^2 - j2y}{(x-1)^2 + y^2} = u + jv$$

w 平面的虚轴:

$$u=0=\frac{x^2+y^2-1}{(x-1)^2+y^2}=0 \Rightarrow x^2+y^2=1 \quad (z \text{ 平面的单位圆})$$

$$z \text{ 平面单位圆} \begin{cases} \text{内} \\ \text{外} \end{cases} \text{ 的点: } x^2+y^2 \begin{cases} < 1 \\ > 1 \end{cases} \Rightarrow \text{在 } w \text{ 平面对应} \begin{cases} u < 0 \\ u > 0 \end{cases}$$